

Notes: Holmström (RES, 1999)

Managerial Incentive Problems: A Dynamic Perspective

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1 Big picture

- **Question.** Can career concerns substitute for explicit incentive contracts, and when do implicit labour-market incentives push managers toward or away from efficient actions?
- **Core mechanism.** A manager's current output affects the market's belief about his ability. Because future wages depend on beliefs, the manager has an implicit incentive to take actions that improve current output and therefore future reputation.
- **Dynamic perspective.** Incentives are intertemporal. The manager may work hard today not because today's wage is contingent on today's output, but because today's output changes tomorrow's perceived ability.
- **Main labour-effort result.** Career concerns can generate effort, but stationary effort is generally below the efficient level when managers discount the future. Efficiency obtains only under restrictive assumptions such as no discounting and sufficiently informative learning.
- **Transient result.** Young managers may overinvest in effort because early output is especially informative when market beliefs about ability are diffuse. This creates a rat-race-like early-career distortion.
- **Technology result.** Nonlinear technology or job assignment can make reputational returns diverge from true productive returns, causing too much or too little effort even with risk neutrality and no discounting.
- **Risk-taking result.** Career concerns can make managers too conservative in investment choice. Risky projects reveal ability, and risk-averse managers may avoid them even when they are value increasing for the firm.
- **Communication result.** If project information is not verifiable, a lemons-type problem can arise: managers may avoid proposing risky projects because investment itself would reveal unfavorable private information about the project pool.
- **Economic takeaway.** Implicit incentives from reputation are real and powerful, but they are not automatically efficient. Career concerns align managerial and firm objectives only when human-capital returns and financial returns are aligned.

2 Incremental contribution of the paper

- **Moves beyond static moral hazard.** Earlier agency models often focused on one-period hidden action and explicit contracts. Holmström studies a dynamic incentive problem in which wages are not explicitly output-contingent, but output affects future market beliefs.
- **Formalizes Fama's market-discipline argument.** Fama (1980) argued that labour-market reputation can police managerial moral hazard. Holmström provides a tractable model showing when that claim is correct and when it fails.
- **Introduces the career-concerns rat race.** The paper shows that early-career incentives can be excessive because early performance has a large effect on market beliefs. This is a dynamic inefficiency, not a static contract-design failure.
- **Separates financial returns from human-capital returns.** A firm's payoff from managerial action need not coincide with the manager's reputational payoff. This distinction is the conceptual core of the paper.
- **Extends career concerns from effort to risk taking.** The paper does not stop at effort supply. It also shows that career concerns can distort project choice and make managers too cautious.
- **Connects career concerns to internal capital budgeting.** The lemons-style example explains why firms may centralize and formalize investment procedures: not only to choose good projects, but also to validate information and evaluate managerial talent.

- **Provides a research agenda.** The conclusion explicitly frames dynamic moral hazard, learning about ability, and investment choice as fertile areas for future work.

Interpretation: Relative to the canonical principal-agent model, the incremental value is to show that incentive problems can arise even without explicit incentive contracts and even when the manager cannot fool the market in equilibrium. The problem is that equilibrium expectations themselves create pressure to perform, because deviations affect the market’s inference about ability.

3 Model objects and measurement

This is a theoretical paper rather than an empirical one. The “data” in the model are the market’s observations of output and the posterior beliefs that those observations generate. The key measured object is the market’s estimate of managerial ability.

3.1 Managerial ability and output

A manager has productive ability, denoted by η . Initially, the market has a prior distribution over ability with mean m_1 and precision h_1 . Precision is the inverse of variance, so higher h means more accurate information. Output in period t is

$$y_t = \eta + a_t + \varepsilon_t, \quad (1)$$

where

- a_t is the manager’s unobserved labour input;
- ε_t is output noise;
- the market observes y_t but not a_t or η separately.

Interpretation: Labour and ability are substitutes in output. A high output can mean high ability, high effort, or good luck. Career concerns arise because the market uses output to update beliefs about ability.

3.2 Preferences and effort cost

The manager is risk neutral in the baseline effort model and has separable intertemporal utility:

$$U(c, a) = \sum_{t \geq 1} \beta^{t-1} [c_t - g(a_t)], \quad (2)$$

where $g(\cdot)$ is increasing and convex.

Interpretation: The manager dislikes effort today but values future wages. If effort improves future beliefs about ability, it has a private reputational return.

3.3 Competitive wage setting

A competitive labour market sets wages equal to expected output conditional on the output history:

$$w_t(y^{t-1}) = \mathbb{E}[y_t \mid y^{t-1}]. \quad (3)$$

Interpretation: The wage is not written as an explicit bonus contract. It is a market wage based on expected productivity. The implicit incentive comes from how current actions affect future expected productivity.

3.4 Belief updating

Let m_t denote the market's posterior mean of ability and let h_t denote posterior precision. When ability is fixed and output noise is normal, Bayesian learning raises precision over time. If ability evolves stochastically, precision may converge to a finite stationary level.

In the stationary ability-shock model, ability follows

$$\eta_{t+1} = \eta_t + \delta_t, \quad (4)$$

where δ_t is an ability shock. The precision dynamics can be expressed with the transformed variable μ_t .

Interpretation: Ability shocks prevent the market from learning ability perfectly. This keeps career concerns alive in the long run, but it also means reputation is an imperfect measure of true productivity.

3.5 Efficient benchmark

The efficient effort level $\bar{a}$ solves

$$g'(\bar{a}) = 1. \quad (5)$$

Interpretation: The firm receives one extra unit of output from one extra unit of effort in the linear technology. Efficiency requires setting marginal effort cost equal to this marginal product.

4 Models and theoretical specifications

4.1 Baseline finite-learning career-concerns model

If ability is fixed, learning eventually becomes complete. The marginal reputational return to effort declines over time because later output is less informative about ability. Holmström shows that equilibrium effort follows first-order conditions of the form

$$\sum_{s=t}^{\infty} \beta^{s-t} \alpha_{ts} = g'(a_t^*), \quad (6)$$

where α_{ts} measures the effect of effort in period t on expected wages in future period s .

Interpretation: Early effort has a larger career value because early output changes beliefs more strongly. As the market learns, reputational incentives fade and effort converges toward zero in the simplest fixed-ability model.

4.2 Stationary model with evolving ability

To sustain career incentives, the paper allows ability to evolve:

$$\eta_{t+1} = \eta_t + \delta_t. \quad (7)$$

The transformed precision variable follows the recursion

$$\mu_{t+1} = \frac{1}{2 + r - \mu_t}, \quad (8)$$

where r is a parameter summarizing the relative precision of output noise and ability shocks.

Stationarity requires

$$\mu^* = 1 + r - \sqrt{r^2 + r}. \quad (9)$$

Interpretation: The fixed point μ^* captures the long-run informativeness of output. Ability shocks add uncertainty, while observed output removes uncertainty. The stationary state balances these two forces.

4.3 Stationary effort

Stationary effort is determined by

$$\beta \frac{\mu^*}{1 - \beta(1 - \mu^*)} = g'(a^*). \quad (10)$$

The efficient benchmark is $g'(\bar{a}) = 1$. Since the left-hand side is typically below one, stationary effort is usually below efficient effort.

Interpretation: Career concerns are generally too weak in the stationary state when managers discount the future. The returns to reputation arrive through future wages, while effort costs are paid immediately.

Model result: Proposition 1

The stationary labour supply a^* is never greater than the efficient level $\bar{a}$. It equals $\bar{a}$ only under restrictive conditions such as no discounting and sufficient noise in both output and ability. Effort is closer to efficient when the discount factor is higher, ability is more stochastic, and output observations are more accurate.

Detailed interpretation: Proposition 1 is the paper's correction to the optimistic version of the career-concerns argument. Reputation can discipline managers, but it does not generally provide the right intensity of incentives.

4.4 Transient dynamics

The key transient question is how effort behaves before the precision of market beliefs reaches its stationary value. If initial precision h_1 is lower than the stationary precision h^* , the market has diffuse information about ability. In that case, early output is very informative.

Model result: Proposition 2

The sequence of optimal labour supply converges monotonically to the stationary level. If initial belief precision is below the stationary level, $h_1 < h^*$, labour supply converges from above. If $h_1 > h^*$, it converges from below.

Detailed interpretation: The empirically natural case is $h_1 < h^*$: young managers are initially poorly known. Their early performance has a large reputational effect, so they may work too hard relative to the long-run career-concerns benchmark.

4.5 Scale economies and nonlinear technology

The linear output technology is special. Holmström next considers nonlinear technology such as

$$y_t = f(\eta_t) + a_t. \quad (11)$$

Because the market infers ability through the nonlinear function $f(\cdot)$, the reputational return to effort need not equal the true productive return to effort.

Interpretation: If ability is convexly rewarded by the labour market, effort can be excessive. If reputational returns are weak relative to productive returns, effort can be too low. The alignment between human-capital returns and financial returns is what matters.

4.6 Risk-taking model

The paper then turns from effort to project choice. A manager’s investment success reveals information about talent. Let π^+ denote the posterior probability that the manager is talented after success and π^- after failure. By Bayes’ rule,

$$\frac{\pi^+}{1 - \pi^+} = \frac{\pi}{1 - \pi} \frac{p_T}{p_U}, \quad \frac{\pi^-}{1 - \pi^-} = \frac{\pi}{1 - \pi} \frac{1 - p_T}{1 - p_U}, \quad (12)$$

where p_T and p_U are success probabilities for talented and untalented managers.

If future wages are linear in posterior beliefs, the expected value of the reputational lottery equals the prior. A risk-neutral manager is indifferent across projects, but a risk-averse manager dislikes the risk in posterior beliefs.

Interpretation: Risky projects expose the manager’s reputation to volatility. Even if the project is valuable for shareholders, a risk-averse manager may prefer safer projects or no project.

4.7 Lemons problem in project choice

Holmström’s final example adds unverifiable private information about project quality. The manager observes a signal about the project, but cannot credibly communicate it. If undertaking the project changes beliefs about both the project and the manager, equilibrium can collapse.

Model result: lemons-style investment collapse

When project information cannot be validated, the only equilibrium in the example is a degenerate equilibrium in which no investments are made. Even a risk-neutral manager behaves conservatively because investing would reveal adverse information.

Interpretation: The result explains why firms may require formal investment procedures, verification, centralized review, and documentation. These institutions do not only control project selection; they also help evaluate managerial talent.

5 Theoretical design and identification logic

This is not an empirical paper, so the relevant “identification” is theoretical: the paper isolates how career concerns create incentives even without explicit performance pay, and then shows which assumptions make those incentives efficient or inefficient.

5.1 Benchmark being tested

The benchmark claim is Fama’s view that competitive managerial labour markets can solve moral hazard by rewarding reputation. Holmström tests this claim by building a model in which:

- wages equal expected productivity;
- expected productivity depends on market beliefs about ability;
- output is informative about ability;

- effort is unobserved and affects output;
- managers care about future wages.

Interpretation: The model gives the market-discipline argument its strongest possible environment: competitive wages, rational beliefs, and no need to fool investors. If inefficiencies arise here, the career-concerns mechanism is not generally efficient.

5.2 What is held fixed

The baseline effort model holds fixed several features to isolate reputation incentives:

- no explicit incentive contract;
- risk-neutral market;
- rational Bayesian updating;
- linear output technology in the benchmark;
- publicly known effort cost function;
- normally distributed noise.

Interpretation: The point is not to match a detailed labour market. The point is to show that even in a clean rational-expectations environment, implicit incentives are distorted by discounting, learning dynamics, and nonlinear technologies.

5.3 Sources of inefficiency

The paper identifies several distinct wedges:

1. **Discounting wedge.** Effort costs are paid today but reputational gains arrive through future wages.
2. **Learning wedge.** Early output is more informative than later output, causing excessive early-career effort and declining incentives over time.
3. **Technology wedge.** Reputational returns to output may not equal the firm's marginal product of effort.
4. **Risk wedge.** Managerial reputation is a risky asset; managers may avoid projects that expose beliefs about ability to downside risk.
5. **Communication wedge.** If project information is not verifiable, investment proposals can suffer from a lemons problem.

Interpretation: The key empirical implication is that observed effort and project choice may be shaped by reputational incentives even when the manager has no explicit bonus contract.

5.4 Why the dynamic structure matters

A static model would miss the main mechanism because career concerns are inherently about the future. Today's action affects tomorrow's belief; tomorrow's belief affects tomorrow's wage. The sequence is:

$$a_t \rightarrow y_t \rightarrow m_{t+1} \rightarrow w_{t+1}. \quad (13)$$

Interpretation: The wage externality is intertemporal. The manager's action changes the market's information set, not the contemporaneous compensation rule.

5.5 Interpretation of propositions as comparative statics

The propositions are comparative statics on implicit incentives:

- higher β makes career concerns stronger;
- more accurate output signals make effort more effective at shaping beliefs;
- more stochastic ability preserves learning and keeps career concerns alive;
- diffuse initial beliefs intensify early-career incentives;
- nonlinear technology and assignment can reverse welfare conclusions.

Interpretation: These comparative statics are the theoretical equivalent of identification: they reveal which primitives drive career-concern incentives and separate the mechanisms from one another.

5.6 Main limitations of the theoretical design

- The model abstracts from explicit contracts, although explicit pay is common in practice.
- The effort model uses risk neutrality and normal learning for tractability.
- The investment examples are stylized and not calibrated.
- The model does not solve the full organizational design problem of combining career concerns, explicit incentives, promotion rules, and internal monitoring.
- The paper raises more questions than it answers about risk taking and communication inside firms.

Interpretation: These limitations are also part of the paper's contribution. Holmström uses simple models to open a dynamic research agenda rather than to provide a final unified theory.

6 Figures and model-result interpretation

6.1 Figure 1: Dynamics of precision and the stationary fixed point

Read:

- The horizontal axis is μ_t , the transformed precision state.
- The vertical axis is μ_{t+1} .
- The curved line is the updating rule $\mu_{t+1} = 1/(2 + r - \mu_t)$.
- The 45-degree line identifies fixed points, where $\mu_{t+1} = \mu_t$.
- The intersection is the unique stationary state μ^* .
- If μ_t starts below the fixed point, the updating path converges upward; if it starts above, it converges downward.

Economic message: The figure shows that the belief-updating system is stable. Ability shocks add uncertainty, output observations remove uncertainty, and their balance creates a stationary precision level.

Detailed interpretation: Figure 1 is not merely a technical diagram. It provides the foundation for studying long-run career concerns. If precision did not converge, the stationary effort analysis would not be well-defined. The fixed point also shows why permanent career-concerns incentives can survive: ability is never fully learned because ability itself evolves over time.

How it fits the paper: The figure prepares the reader for Proposition 2. Once the precision state converges monotonically, the manager's effort incentives can also be shown to converge monotonically toward the stationary effort level.

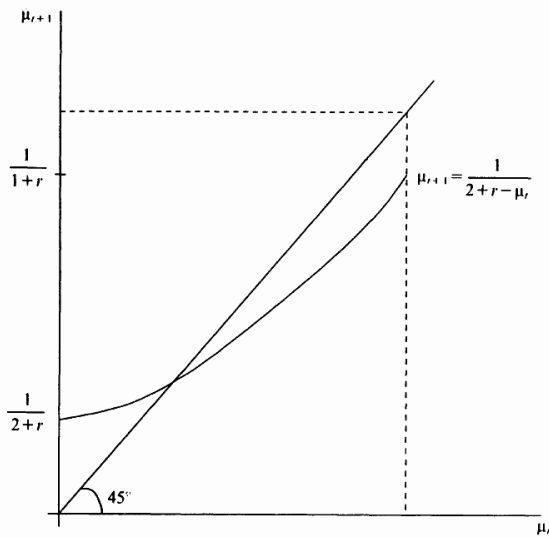


FIGURE 1

Figure 1: Dynamics of precision. The original figure plots the precision updating map against the 45-degree line. The stable fixed point captures the stationary amount of information the market has about ability when output observations and ability shocks interact.

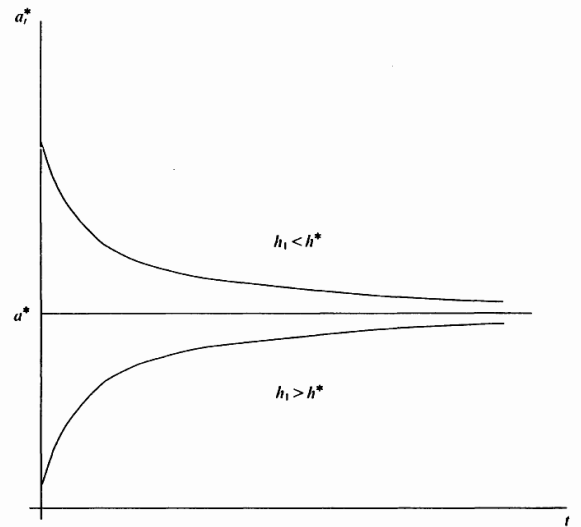


FIGURE 2

Figure 2: Convergence of labour supply. The original figure shows how optimal labour supply moves toward the stationary level. Initial uncertainty about ability determines whether effort begins above or below the stationary benchmark.

6.2 Figure 2: Convergence of labour supply to the stationary level

Read:

- The horizontal axis is time t .
- The vertical axis is optimal labour supply a_t^* .
- The horizontal line is stationary labour supply a^* .
- If initial precision is low, $h_1 < h^*$, effort starts above the stationary level and converges downward.
- If initial precision is high, $h_1 > h^*$, effort starts below the stationary level and converges upward.
- The empirically natural case is $h_1 < h^*$: early-career ability is initially imprecisely known.

Economic message: The figure illustrates the rat-race implication of career concerns. Young or newly appointed managers may overwork because early output has the largest reputational effect.

Detailed interpretation: Figure 2 is the most intuitive diagram in the paper. It links the abstract precision dynamics to observed managerial behaviour. If the market begins with diffuse information about a manager, effort is front-loaded. This can look productive in the short run, but it is not necessarily efficient because the incentive is driven by reputation rather than by the contemporaneous marginal product of effort.

How it fits the paper: The figure supports the broader message that implicit incentives are dynamic and history dependent. The same manager facing the same technology may choose different effort levels at different career stages because the informativeness of performance differs over time.

6.3 Model-result box: Proposition 1, stationary effort

Read:

- Stationary effort is bounded above by efficient effort.
- With discounting, career concerns are too weak in the long run.
- More accurate output observations raise reputational incentives.
- More stochastic ability raises long-run learning value and helps sustain incentives.

Economic message: Reputation can discipline managers, but it is not a free substitute for explicit contracts. The labour market's implicit incentives typically underprovide long-run effort.

6.4 Model-result box: Proposition 2, transient effort

Read:

- Effort converges monotonically to the stationary level.
- Low initial precision generates overinvestment in effort.
- High initial precision generates underinvestment relative to the stationary level.
- The young-manager case is associated with low initial precision and therefore high initial career-concerns effort.

Economic message: The dynamic model predicts excessive early-career effort even without explicit tournaments or promotion contests. Reputation alone can create a rat race.

6.5 Model-result box: Risk taking and lemons problem

Read:

- Investment outcomes reveal managerial talent.
- Risk-averse managers dislike reputational risk and may avoid value-increasing risky projects.
- If project signals are private and unverifiable, proposing a project can reveal bad information.
- In the extreme example, the equilibrium collapses to no investment.

Economic message: Career concerns can produce conservatism in project choice. This helps explain why firms may need internal screening, centralized capital budgeting, and project-validation procedures.

7 Short summary

- The paper studies managerial incentives when wages are set by labour-market beliefs about ability.
- Managers care about future wages, and future wages depend on current output through reputation.
- The baseline model formalizes Fama’s idea that career concerns can discipline moral hazard.
- Holmström shows that the idea is only partly right.
- Stationary effort is generally below the efficient level when managers discount future reputational gains.
- Early-career effort can be above the stationary level when initial ability is imprecisely known.
- Nonlinear technologies and job assignments can make reputational returns diverge from productive returns.
- Career concerns also affect risk taking, not only effort.
- Risk-averse managers may avoid risky projects because investment outcomes reveal ability.
- If project information is unverifiable, a lemons problem can lead to excessive conservatism.
- The central lesson is that implicit incentives depend on the alignment between human-capital returns and financial returns.

8 Conclusion

8.1 Core conclusion

Holmström’s core conclusion is that career concerns are a powerful but imperfect incentive mechanism. They can motivate managers even without explicit output-contingent contracts, because output affects future perceptions of ability. But this same reputational force can generate inefficient effort, excessive early-career work, and conservative investment behaviour.

8.2 Economic intuition

The intuition is that managers do not maximize the firm’s financial returns directly. They maximize a combination of current wages, effort costs, and future human-capital returns. If the market interprets today’s output as a signal of ability, managers will take actions that improve that signal. Those actions may be productive, but they may also be distorted if the market’s inference problem differs from the firm’s profit problem.

8.3 Why the paper matters

The paper matters because it reframes managerial incentives as a dynamic learning problem. It shows that labour markets can provide incentives, but only through a noisy process of belief updating. This perspective helps explain why managers may work especially hard early in their careers, why they may avoid risky projects, and why organizations use internal procedures to evaluate projects and managers.

8.4 Incremental contribution in one sentence

The paper’s incremental contribution is to show that reputation-based incentives are not automatically efficient: they depend on the dynamic mapping from performance to beliefs, from beliefs to wages, and from wages to managerial choices.

8.5 Policy and organizational implications

- Firms should not assume that market discipline alone solves agency problems.
- Explicit contracts may be needed when career concerns are too weak or misaligned.
- Young managers may face excessive reputational pressure, creating inefficient overwork or short-termism.
- Risk-taking procedures should account for managers' concern about how failure affects reputation.
- Centralized capital budgeting and project documentation can be interpreted as mechanisms for validating information, not merely as bureaucratic control.

8.6 Limitations and open questions

The paper deliberately uses stylized models. It does not fully solve how firms should optimally combine explicit contracts, promotions, job assignments, monitoring, and reputation. It also does not empirically test the model. Instead, it identifies mechanisms that later empirical and theoretical work can study: early-career rat races, reputation-driven conservatism, and the role of organizations in validating managerial information.

8.7 Takeaway

Career concerns are double-edged. They can motivate effort when explicit incentives are absent, but they can also distort behaviour because the manager is investing in reputation rather than directly maximizing firm value. The central question is whether reputational returns and productive returns are aligned. When they are not, implicit incentives can be as much a source of inefficiency as a solution to moral hazard.